SESSION 2

CQs:

1. **CQ21:** What are the pixel dimensions (mtsd:width and mtsd:height) of the image sourced from the file --7fWq6WjZM8LleUSuvOEA.json?
2. **CQ22:** Which specific mtsd:TrafficSignObject entities does a given image contain via the mtsd:hasObject property?
3. **CQ23:** Is a specific image flagged as a panoramic view (mtsd:isPano is true)?
4. **CQ24:** What are the exact bounding box coordinates (mtsd:xmin, mtsd:ymin, mtsd:xmax, mtsd:ymax) for the traffic sign object with the key iw8oed2vtpa46rfvmki2wk?
5. **CQ25:** Can the system retrieve all traffic sign objects in the dataset where the physical bounding box property mtsp:occluded is marked as true?
6. **CQ26:** Which Mapillary image keys (mtsd:key) contain a sign identified as MTSD_Regulatory_GiveWayToOncomingTraffic_G1?
7. **CQ27:** How many total images in the dataset feature a sign that is logically equivalent to the Vienna Convention's PriorityForOncomingTraffic class?
8. **CQ28:** If a mtsd:TrafficSignObject has an rdfs:label of regulatory--bicycles-only--g1, what is its mandatory ground colour according to its equivalent Vienna Convention class?

LLM: Gemini

LLM Prompt: Role: You are an expert Ontology Engineer specializing in ontology alignment and dataset mapping.
Objective: Generate OWL2 RDF/XML code to create classes for a provided list of Mapillary Traffic Sign Dataset (MTSD) labels and map them to their equivalent Vienna Convention concepts using owl:equivalentClass.
Scope & Rules:
Input: I will provide a list of string labels from the MTSD (e.g., regulatory--end-of-priority-road--g1).
Class Creation: For each label, create an <owl:Class>. The URI should use the base http://example.org/mtsd/class# and convert the label string into a clean PascalCase or camelCase ID (e.g., MTSD_Regulatory_EndOfPriorityRoad).
Equivalence Mapping: You must deduce which Vienna Convention sign this MTSD class corresponds to. Map it using <owl:equivalentClass>.
Target Namespace: Assume my Vienna Convention base URI is http://www.ontology.org/traffic/vienna-convention#.
Output Format: Provide ONLY the raw RDF/XML code snippet containing the new class declarations and their equivalence axioms. Do not write explanations.
Here is the batch of MTSD labels to process:
- regulatory--bicycles-only--g1
- regulatory--bicycles-only--g2
- regulatory--bicycles-only--g3
- regulatory--buses-only--g1
- regulatory--detour-left--g1
- regulatory--do-not-block-intersection--g1
- regulatory--do-not-stop-on-tracks--g1
- regulatory--dual-lanes-go-straight-on-left--g1
- regulatory--dual-lanes-go-straight-on-right--g1
- regulatory--dual-lanes-turn-left-no-u-turn--g1
- regulatory--dual-lanes-turn-left-or-straight--g1
- regulatory--dual-lanes-turn-right-or-straight--g1
- regulatory--dual-path-bicycles-and-pedestrians--g1
- regulatory--dual-path-pedestrians-and-bicycles--g1

- regulatory--end-of-bicycles-only--g1
- regulatory--end-of-buses-only--g1
- regulatory--end-of-maximum-speed-limit-30--g2
- regulatory--end-of-maximum-speed-limit-70--g1
- regulatory--end-of-maximum-speed-limit-70--g2
- regulatory--end-of-no-parking--g1
- regulatory--end-of-priority-road--g1
- regulatory--end-of-prohibition--g1
- regulatory--end-of-speed-limit-zone--g1
- regulatory--give-way-to-oncoming-traffic--g1
- regulatory--go-straight--g1
- regulatory--go-straight--g3